Waveshare Staircase Panel

Visual Install Guide - 6 steps

This guide walks the carpenter, electrician and homeowner through installing the Waveshare 10.1" touch screen on the staircase wall. Six numbered pictures take you from the bare wall to a finished panel that pulls out for service.

How it works: the home server (MSI Cubi 5) drives the screen. Two cables run up the existing 25 mm conduit from the server: HDMI for the picture (from the server's second screen output) and USB for the touch. There is no computer at the screen. The screen does NOT take power from the USB - it has its own 12V power socket, so 12V is provided at the wall. An NFC reader hidden behind the lower-right bezel lets you tap a phone or tag to start a scene. The walnut frame holds on with 4 magnets, so the whole panel pulls out for service without damaging the wall.

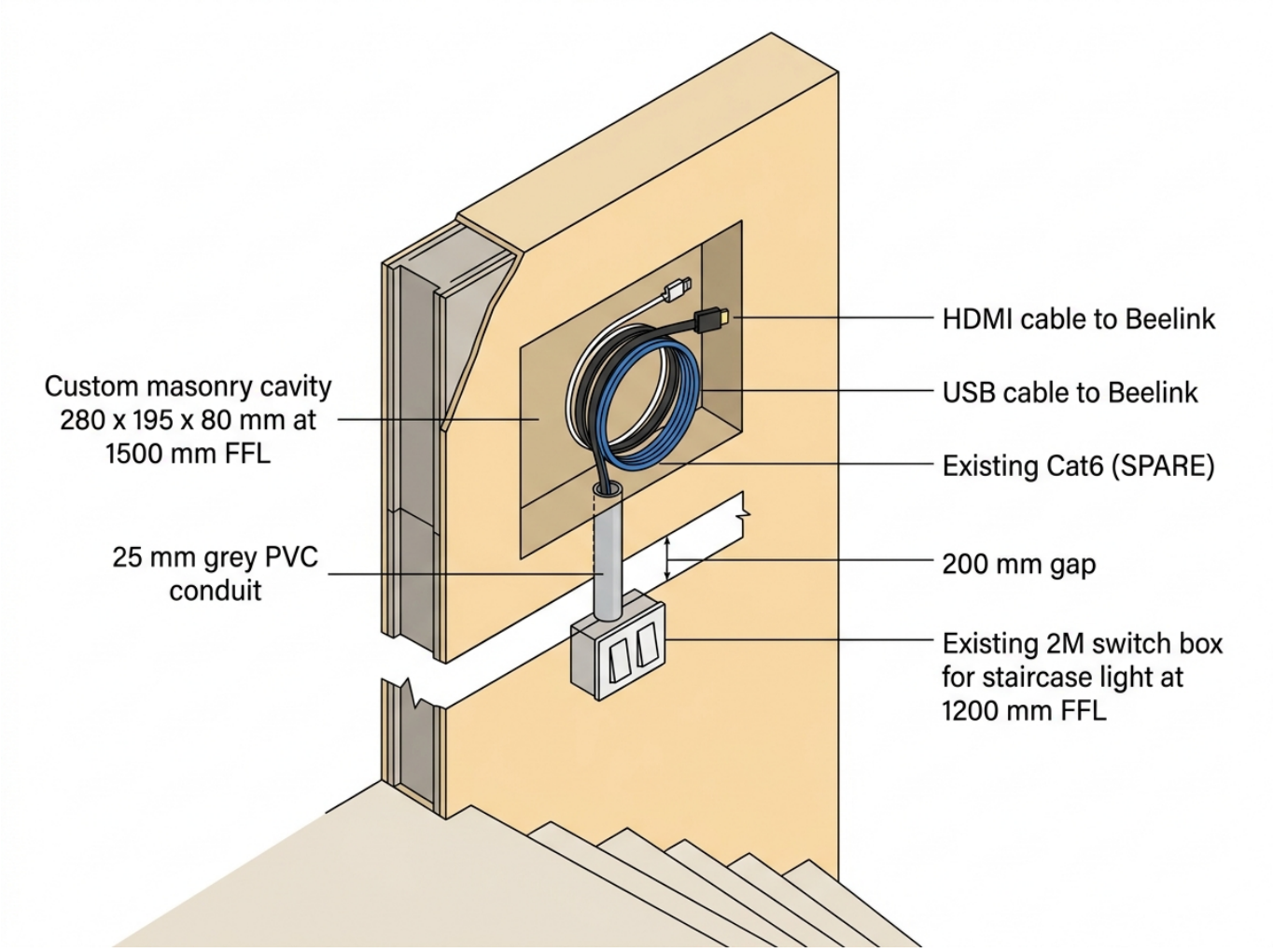
Before you start - 4 things to know

1. POWER IS 12V, NOT 5V. The screen has its own 12V power socket (12V, 2A). The USB cable carries touch only - it does not power the screen.
2. THREE PLUGS at the screen: HDMI + touch USB + 12V power.
3. USE L-SHAPED (90 degree) PLUGS. The sockets sit in a recess on the back of the screen - straight plugs will not fit.
4. TIGHT FIT. Only about 6-8 mm spare around the screen in the 280 x 195 mm cavity. The carpenter builds the frame AFTER the screen arrives, around the real screen.

Quick reference

Location:	Staircase South wall, 1500 mm FFL, ~200 mm above the existing 2M light switch
Cavity:	280 x 195 x 80 mm masonry cavity (no back box)
Screen:	Waveshare 10.1" HD touch LCD (G) with case, 1920 x 1200. Case 263.45 x 182.45 x 23.6 mm; picture 217.91 x 136.50 mm; 4x M4 screw holes on the back (75 mm square, VESA 75)
Buy from:	ThinkRobotics (the first order, at ElectroPi, was cancelled)
Power:	12V, 2A into the screen's own power socket - NOT 5V, NOT over USB. (a) PoE splitter on the Cat6 (preferred) or (b) a small 12V supply off the staircase switch 230V
Plugs:	3 at the screen - HDMI + touch micro-USB + 12V - all L-shaped (90 degree)
Conduit:	D5 on the map (C-Server-Waveshare, 25 mm grey, ~2 m): HDMI + touch USB + Cat6
Server:	MSI Cubi 5 - its second screen output (HDMI) + one USB port
Mount:	Walnut bezel + 4-magnet service hatch, built around the real screen
Full spec:	electrical/WAVESHARE_INDOOR_PANEL.md (v3.3)

Step 1 - The wall cavity before install - what comes into it



PICTURE NOTE: This picture is from the old plan. Where it says 'Beelink', read 'server' (now the MSI Cubi 5). The Cat6 is NOT spare any more - it is the preferred 12V power feed, option (a). The picture does not show the 12V power - follow the text below.

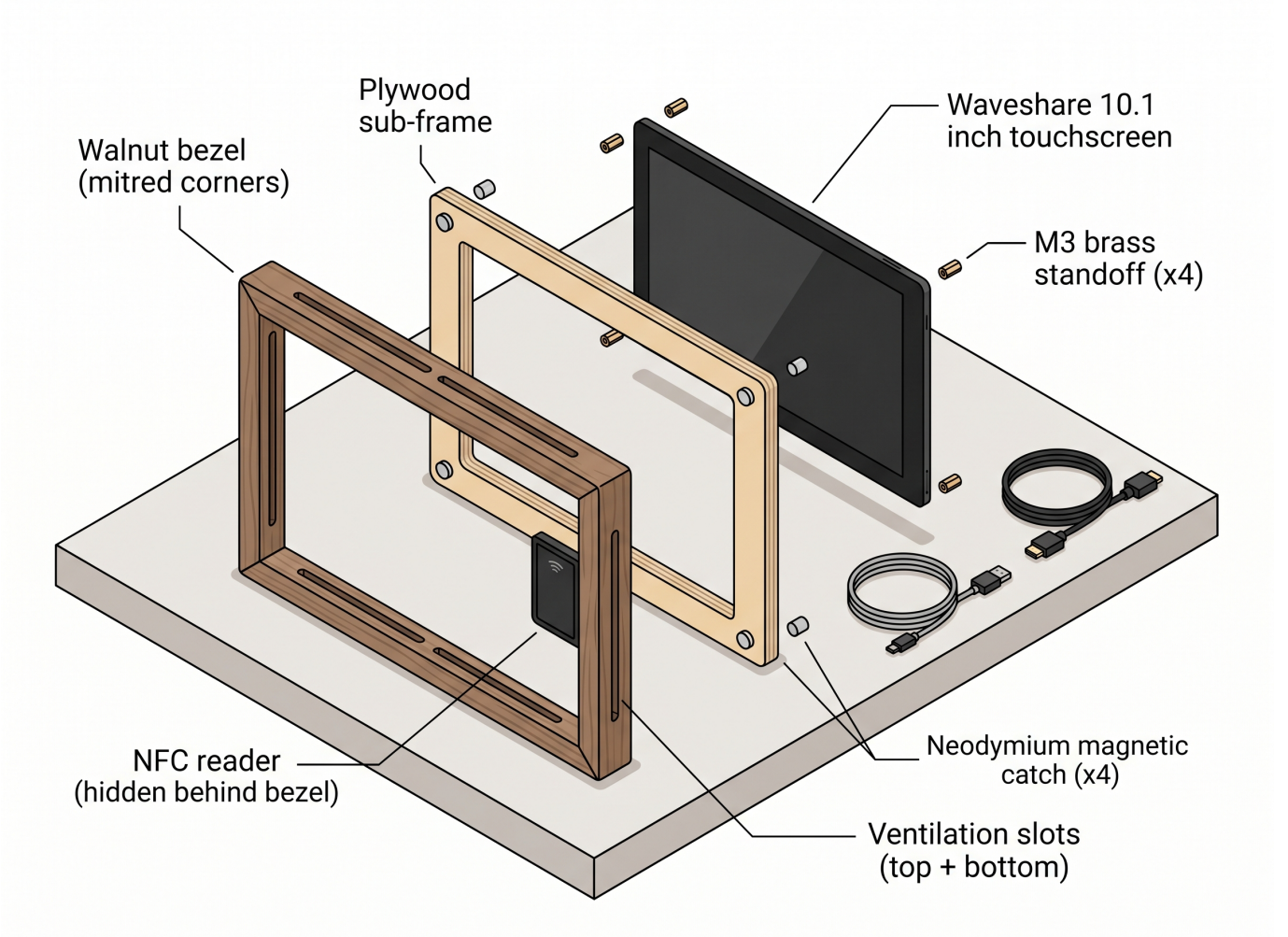
A 280 x 195 x 80 mm cavity is cut into the 9-inch staircase wall, centred at 1500 mm FFL, about 200 mm above the existing 2M staircase light switch. There is no back box - it is a plain masonry cavity. The 25 mm grey conduit comes in at the bottom with three cables from the server: HDMI (the picture), USB (the touch only) and the Cat6 that is already there. Leave a 200 mm service loop of every cable, including the 12V lead, coiled inside the cavity - later the screen must pull out 150 mm for service.

POWER: the screen needs its own 12V, 2A. Pick one way on site:

(a) PREFERRED - send power up the Cat6 from a PoE port at the server. A small PoE splitter box in the cavity (it takes power from the network cable and gives out 12V) feeds the screen.

(b) A small 12V power supply in or next to the staircase switch box, fed from the switch's 230V, with a short 12V cable up to the cavity. The 230V stays at the switch box. With (b), the Cat6 stays spare: fit an RJ45 plug and label it 'SPARE'.

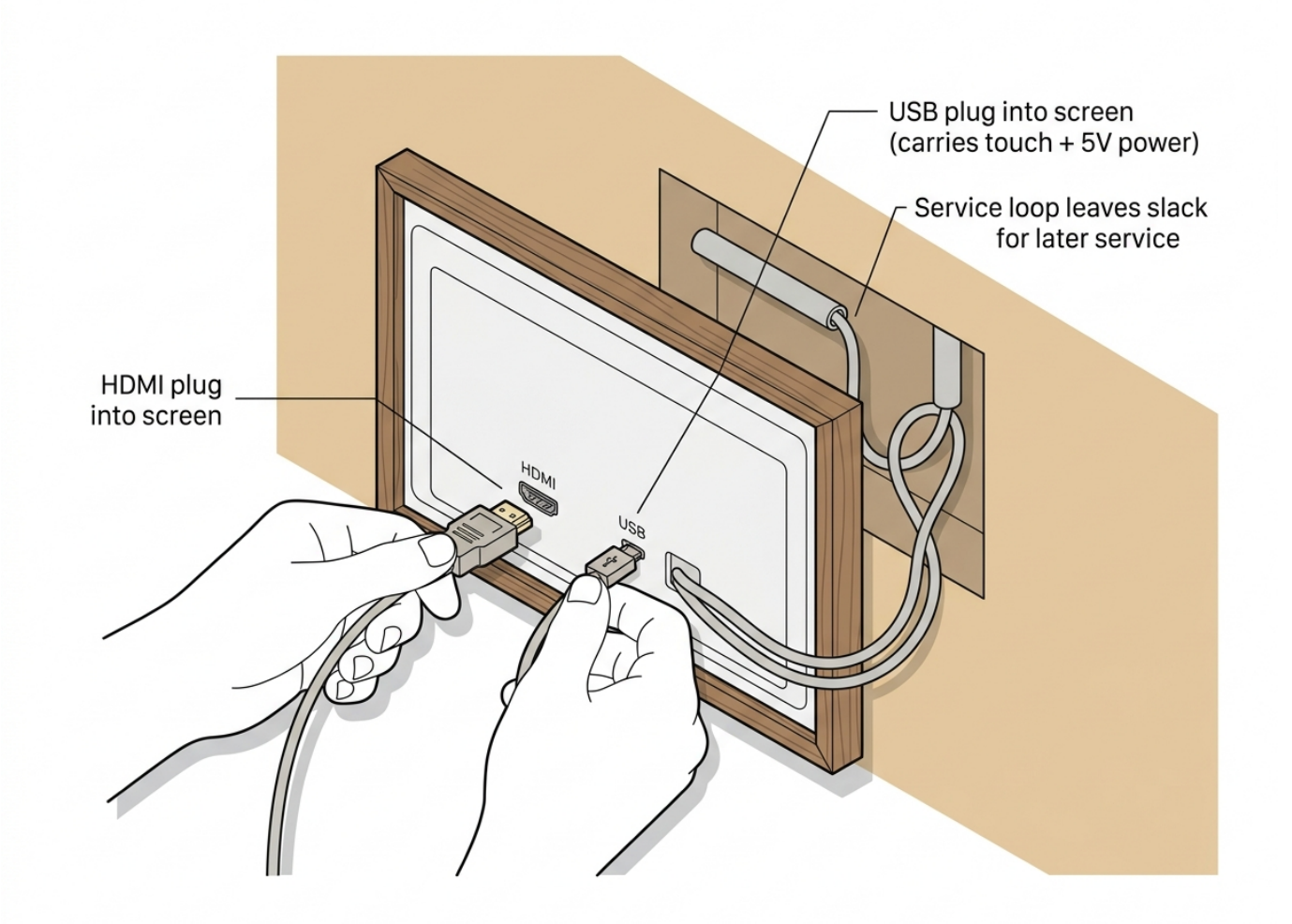
Step 2 - Bench assembly - build the frame around the real screen



PICTURE NOTE: The picture shows small M3 standoffs at the screen corners. The (G) screen has 4 M4 screw holes in a 75 mm square on its back - fix it with those. The picture also shows only 2 cables; there is a third, the 12V power lead.

Wait until the screen arrives before the carpenter builds anything - the fit is tight. The (G) screen in its case is 263.45 x 182.45 x 23.6 mm, so the 280 x 195 mm cavity leaves only about 6-8 mm spare around it. First re-measure the real cavity (masonry is never exact), then build the walnut bezel and plywood sub-frame around the actual screen: 4 magnet catches at the corners, vent slots top and bottom, and a 220 x 140 mm window for the picture (the picture itself is 217.91 x 136.50 mm). The back of the screen has 4 M4 screw holes in a 75 mm square (VESA 75) - use these to fix it to the sub-frame; the carpenter decides how once the screen is in hand. Stick the ACR122U NFC reader inside the lower-right bezel - it reads taps through the wood.

Step 3 - Connecting the 3 cables to the back of the screen



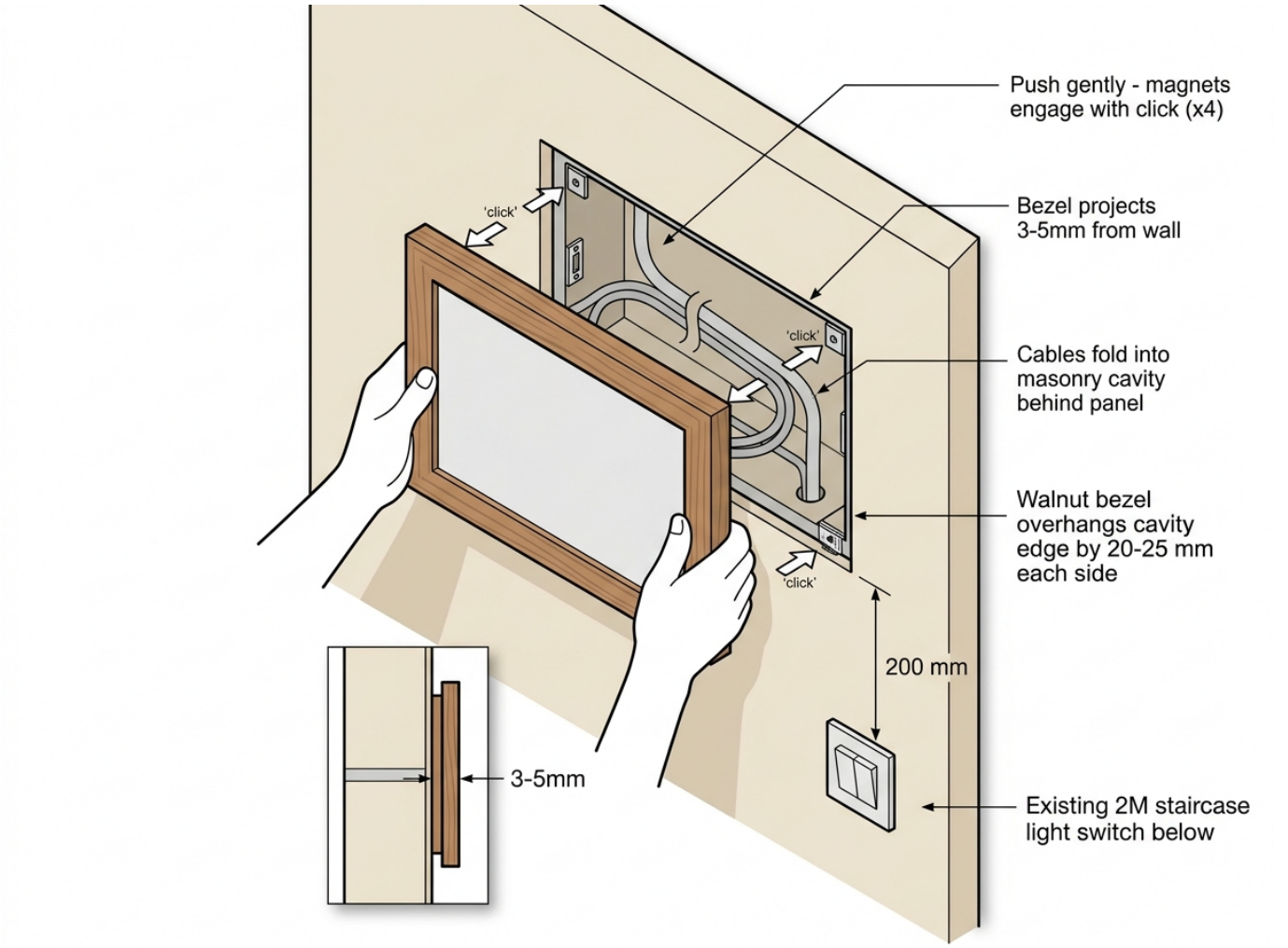
PICTURE NOTE: This picture is from the old plan and is WRONG about power. It says the USB carries 5V power - it does not. The USB carries touch only. There are 3 plugs, not 2: HDMI + touch USB + a separate 12V power plug, all L-shaped.

Pull the cables out of the cavity (the service loop gives you slack) and plug them into the back of the screen. There are 3 plugs:

1. HDMI - the picture, from the server.
2. Touch USB (micro-USB) - touch only. It does NOT power the screen.
3. 12V power (round DC plug) - from the PoE splitter or the 12V supply. 12V, 2A - NOT 5V.

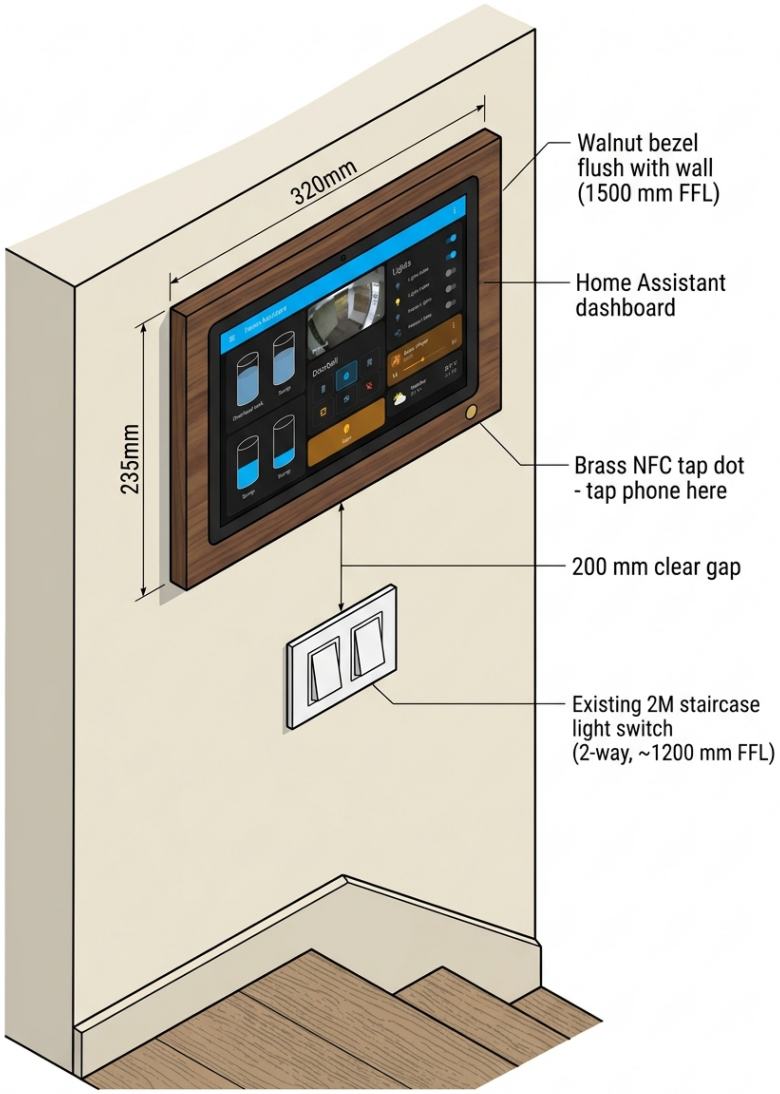
The sockets sit in a recess on the back and the plugs go in sideways, so use L-shaped (90 degree) plugs or adapters on all 3 - straight plugs will not fit in the cavity. Check that the 12V plug fits the screen's power socket (compare it with the 12V adapter in the screen's box). Leave the socket marked '5V 3A' empty - it gives power OUT (for a Raspberry Pi); it does not power the screen. Don't fix anything yet - you push the whole unit into the wall next.

Step 4 - Pushing the assembly into the wall cavity - magnets engage



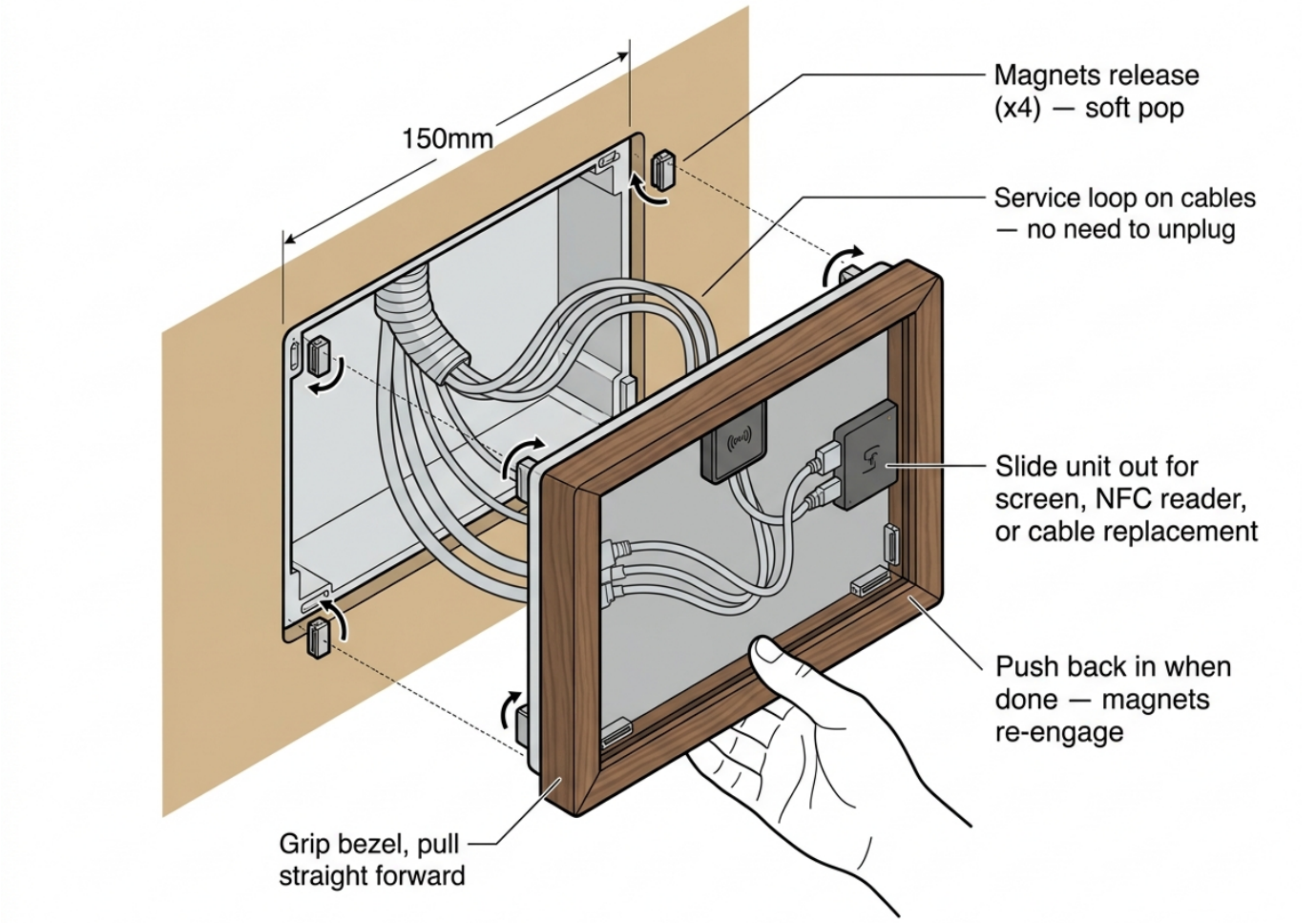
Hold the walnut bezel flat against the wall, lined up with the cavity opening, and push gently. Feed the cables (and the PoE splitter box, if you used option a) into the cavity behind the screen without pinching them - there is only about 6-8 mm spare around the screen. The four magnets in the corners catch onto steel strike plates fixed inside the cavity with a soft click. The bezel sits flush around the cavity opening, projecting only 3-5 mm from the wall - a subtle shadow line, not a tablet bolted on the wall - and overlaps the wall by 20-25 mm on every side to hide the cut edge. Below the new screen, the existing 2M staircase light switch is vertically aligned with the cavity.

Step 5 - Finished install - screen showing Home Assistant dashboard



The completed staircase panel: walnut bezel sitting flush with the wall at 1500 mm FFL, and the sharp 1920 x 1200 screen showing the Home Assistant dashboard (water tank levels, light controls, doorbell tile, music, weather). The small brass 'tap here' dot in the lower-right corner of the bezel marks where to tap a phone or NFC sticker to trigger a scene. Vertically aligned below the screen with about 200 mm gap is the existing 2-module switch for the staircase light (a 2-way switch - same control point as the top of the stairs), creating a clean vertical control stack near the bottom of the staircase.

Step 6 - Service mode - pulling the panel forward for maintenance



If anything inside needs service - screen replacement, cable swap, NFC reader change - grip the bezel and pull straight out. Magnets release with a soft pop, and the whole assembly slides forward about 150 mm on the cable service loop (HDMI, touch USB and the 12V lead all need that slack). No need to unplug anything. This is also how you reach the brightness buttons on the back of the screen. When done, push back in - magnets catch, click into place, flush again. No wall damage, no tools.

Reference and next steps

Full hardware + mount spec:
electrical/WAVESHARE_INDOOR_PANEL.md v3.3 (15-section site handout). Where it says 'Beelink', read 'the server' - now the MSI Cubi 5.

Related conduit documentation:
electrical/conduits-and-cavities.md (C-Niche-Waveshare row) = D5 / C-Server-Waveshare on the conduit map

Decision history (decisions/decision-log.md):
2026-05-25 - hybrid panel design (staircase screen driven straight from the server)
2026-06-02 - server changed to the MSI Cubi 5; screen changed to the (G) 1920 x 1200
2026-09-28 - ElectroPi cancelled the (G) order; the same (G) re-chosen, to buy from ThinkRobotics; power corrected to 12V, 2A

Parts for the power and plugs:
(a) PoE splitter: ask the shop for a 'PoE+ (802.3at) splitter, 12V 2A output' with a DC plug that fits the screen - or (b) a small 12V 2A power supply.
Plus L-shaped (90 degree) HDMI, micro-USB and DC plugs or adapters for the screen end.

Homeowner checks before buying:
- Confirm the MSI Cubi 5 model has a second screen output free for this panel (HDMI, or USB-C with an HDMI adapter).
- WhatsApp ThinkRobotics to confirm the (G) is really in stock before paying (backup shop: REES52).
- If you choose power option (a), get the bigger PoE switch (TP-Link Omada ES220GP) - the splitter adds up to ~25 W of load.

Next session - once panel is installed:
Home Assistant dashboard layout (separate session, live editing on the actual panel)

Dining hall companion panel (future):
Cat6 stub already provisioned via C-Niche-Dining conduit. Architecture: Pi-at-screen wired, matching the foyer welcome system pattern. Install when interior designer's cantilever shelf design is finalised.